Enhancement Three Narrative

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For the algorithms and data structures enhancement, I selected the Corner Grocer application. I developed this C++ program for my final project in the SNHU Programming Languages course, which I took two years ago. In the original version of this application, the program would begin by reading a text file containing purchase history data for the produce section of Corner Grocer. The program would load the name of each unique item into a vector and load the name of each unique item and how many times it had been purchased into a map. By interacting with a menu, the user could accomplish two tasks: view how many times a particular produce item was purchased or view a complete list of how many times each produce item was purchased. When viewing the complete list, the user could choose whether the number of times each item was purchased would be represented by digits or a histogram.

I chose to include this artifact in my portfolio because a real-world version of this application would use a database. Using a file to store purchase history data indicates that the current process of recording and/or obtaining this data is inefficient. Whether users are manually hand-jamming purchase history data into this file as customers make purchases or if they are downloading a copy of the data from a second storage location, neither of these methods are efficient for a business because they involve unnecessary manual steps. Storing purchase history data in a database is a better long-term solution for this application for many reasons. First, it will ensure that there is a single source of truth. Storing large amounts of ever-changing data in a file gets messy because there could be multiple versions of a file, making it difficult to identify which file has the most up-to-date data. Second, using a database is a more secure method of data storage; it is hard to ensure the security of a file because it is very easy to create copies and send them to anybody. Third, using a database will allow for scalability in the future. Databases are easy to scale up because they can handle large amounts of data without huge impacts to performance, whereas a file containing purchase history information for a business would very quickly have negative impacts to performance and overwhelm users. For all these reasons, this Python program will be greatly improved by connecting directly to a database storing purchase history information.

I chose to use MongoDB for the Corner Grocer's sales database because its non-relational structure provides flexibility and scalability. I chose to use MongoDB because I knew I could utilize the PyMongo driver to easily integrate the database with the Python program developed in the software engineering and design enhancement, and I wanted to gain more experience with non-relational databases. I utilized MongoDB Atlas to deploy my database. This cloud service provides preconfigured security features such as encryption in transit and at rest (MongoDB, n.d.). I also defined my own user roles that require a username and password (which are stored securely in MongoDB Atlas and are not hardcoded in the Python program) for authorization and authentication. Users of the program I developed are required to enter a valid username and password before they can query the database.

I created my own document structure that included the object id, the item name, the number of items purchased, and the category. I created the category field so Corner Grocer could choose to expand this database and create additional categories such as dairy, canned goods, and bakery. To populate records in the database, I used ChatGPT to generate sample data records that aligned with my document structure. By creating this database, I showcased my ability to design and initialize a secure, non-relational database. I also demonstrated my ability to write database queries and create a cohesive, scalable software application designed for long-term use.

I did meet the course outcomes that I planned to meet for the database enhancement and met an additional one. I met course outcome four by showcasing my ability to implement features that add value to this application by creating a secure database to store Corner Grocer's purchase history data, providing a more robust, long-term solution. I also met course outcome five by utilizing the built-in security features provided by MongoDB Atlas and by creating appropriate user roles to authenticate and authorize users. Finally, my in-line comments and this narrative demonstrate my ability to develop and deliver professional written communication, which shows progress in meeting course outcome two.

As I was working on this enhancement, I solidified my understanding of MongoDB and its non-relational structure. I made sure to learn how to interact with the database using as many methods as I could, including the mongo shell, the built-in MongoDB extension for Visual Studio Code, and the MongoDB Atlas data explorer. In the end, my biggest takeaway was that the data explorer helped me better visualize my cluster and assisted me with determining the correct commands to use in the mongo shell and built-in MongoDB extension for Visual Studio Code.

Utilizing the MongoDB documentation for creating and initially connecting to my database was a great resource for me, especially when I faced challenges. For example, I initially had trouble creating a successful connection to my database. I had to try a handful of different methods before I was able to make a successful connection. In the end, I was able to connect after adjusting my connection string to align with the structure found in the MongoDB documentation and resetting my password.

References

MongoDB, Inc. (n.d.). Configure Security Features for Clusters. MongoDB Docs. https://www.mongodb.com/docs/atlas/setup-cluster-security/